

Pooja Manjunatha

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PROFESSIONAL EXPERIENCE

Virufy, San Jose | *Software Engineer - Machine Learning*

Feb 2025- Present

- Designed and implemented **automated Python pipelines** for multi-source audio data, delivering **production-ready inputs** to downstream systems.
- Orchestrated **containerized applications** using **Docker on AWS EC2**, enabling **scalable and reproducible deployments**.
- Built and deployed **REST APIs** with **FastAPI and Flask**, exposing ML services for real-time use in production.
- Developed **SQL-based backend services** and optimized queries, improving **processing efficiency by 20%**.
- Led **code reviews and agile sprint planning**, improving **code quality and cross-team collaboration**.
- Applied **OOP-based modular design**, boosting **reusability and maintainability** across services.
- Integrated **AWS Glue and S3 pipelines on Red Hat** to enforce **schema compliance, data validation, and security requirements (HIPAA)**

Reinfosec, India | *Data Analyst*

Aug 2021-Apr 2022

- Maintained **cloud-based ETL pipelines** with **Python and GCP Cloud Functions** to ingest and normalize security logs into **Snowflake**.
- Implemented **SQL-driven risk scoring algorithms** to enrich alerts with contextual metadata, **reducing low-priority alerts by 40%**.
- Developed **interactive Tableau dashboards** to track trends, SLA breaches, and team performance, enabling faster anomaly detection and **improving data-driven operational decisions**.

Apexon, India | *Data Engineer Intern*

Jun 2023-Aug 2023

- Designed and implemented **ETL pipelines using PySpark and SQL** for HR support ticket data.
- Automated reporting and analytics workflows using **Azure Databricks**, reducing manual effort and improving accuracy.
- Developed **Python scripts** to extract, clean, and transform **structured and unstructured data**.

Luceats Consulting, India | *Software Developer Intern - Machine Learning*

Apr 2022 – Jul 2022

- Preprocessed **unstructured vendor invoice data** using OCR and **structured pipelines for downstream analytics** using Python.
- Integrated multiple internal and external data sources, improving extraction accuracy and consistency.
- Designed **rule-based recommendation system** for extraction pipelines, optimizing **operational efficiency**.

SKILLS

Programming: Python (3.x), Java, JavaScript, R, SQL

Web & API Development: Django, Flask, FastAPI, RESTful APIs, GraphQL, OAuth, React

Cloud & DevOps: AWS, GCP, Azure, Docker, Kubernetes, Terraform, Ansible, CloudFormation

CI/CD & Version Control: Jenkins, GitLab CI, CircleCI, Git, GitHub, GitLab

Databases & Data Engineering: MySQL, PostgreSQL, MongoDB, Snowflake, BigQuery, Apache Airflow, Luigi, Prefect

Data Science & ML: Pandas, NumPy, PySpark, Dask, Scikit-learn, TensorFlow, PyTorch

Testing & Visualization: PyTest, unittest, Selenium, Tableau, Power BI, Matplotlib, Seaborn

Other: Linux, macOS, Windows, Agile (Scrum/Kanban), Jira, Trello, SDLC

EDUCATION

M.S. Data Analytics – San José State University, San Jose, CA

Jan 2023–Dec 2024

M.S. Business Analytics – The University of Texas at Dallas, Dallas, TX

Aug 2022–Dec 2022

B.Tech, Information Science – Vemana Institute of Technology, India

Aug 2017–May 2021

PROJECTS

• Kafka on Kubernetes

Deployed **multi-broker Kafka clusters** with **StatefulSets and custom manifests**, enabling reliable and scalable streaming. Automated deployments, improving **fault-tolerance and pipeline stability**

• Recommendation System with Gemini, GraphRAG & Knowledge Graphs

Integrated Google Gemini LLM with GraphRAG and ontology knowledge graphs. Orchestrated ETL workflows in GCP Dataflow and BigQuery to process and serve large-scale user and product interaction data. Boosted recommendation relevance by ~30%.

• Real-Time Stock Market Data Pipeline

Built streaming pipelines using Apache Kafka and AWS Glue for real-time ingestion and transformation of high-frequency market data. Implemented analytical data marts in Amazon Athena.

• Gun Violence Data Analysis & ETL

Normalized 1M+ incident records and deployed ETL pipelines to Snowflake, reducing query latency by 50%.

• NYC Parking Violations Modeling with dbt

Designed end-to-end dbt pipeline on DuckDB with bronze/silver/gold layers. Implemented 15+ automated data quality checks, reducing inconsistencies by 90%.

• Beauty Product Recommendation Engine

Built ensemble model using LightFM + NLP embeddings. Developed ETL pipelines for structured and unstructured data, achieving 74% accuracy.